

DIVYANSHU TANDON

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EDUCATION

Arizona State University (ASU), Tempe, Arizona

May 2027

Bachelor of Science in Computer Science

- ♦ Computer Systems Architecture
- ♦ Data Structures and Algorithms
- ♦ Applied Foundations of Cybersecurity
- ♦ Probability and Statistics
- ♦ Object-Oriented Software Designs
- ♦ Operating Systems

SKILLS

Python, Java, C, C++, JavaScript, TypeScript, MySQL, HTML/CSS, React, FastAPI, REST APIs, NumPy, pandas, h5py, Linux, PyQt / PySide, Bash / Shell Scripting, Git, GitHub, GitLab

WORK EXPERIENCE

Research Assistant, BDI Biodesign - Center for Applied Structural Discovery, ASU

Sept 2025 - Present

- ♦ Architected Python pipelines to process high-throughput liquid microjet image datasets for experimental evaluation.
- ♦ Calibrated automated instrumentation setups to streamline high-speed data acquisition.
- ♦ Coordinated lab logistics and operational workflows across active engineering experiments.

Undergraduate Researcher, CXFEL (Compact X-ray Free Electron Laser), ASU

May 2024 - Present

- ♦ Developed Python desktop tools using NumPy and PyQt to optimize biomolecular imaging software.
- ♦ Contributed to data analysis and automation pipelines supporting research at the Linac Coherent Light Source (LCLS) and ASU Compact X-ray Light Source (CXLS).
- ♦ Implemented matrix and image-processing algorithms to eliminate computational bottlenecks in diffractive imaging.

Summer Researcher, LCLS(Linac Coherent Light Source), SLAC National Accelerator Laboratory

July 2025

- ♦ Operated and optimized ultra-thin sheet-jet injectors for in-solution Single-Particle Imaging (SPI) of high-concentration biological macromolecules, achieving stable delivery and minimal background scattering.
- ♦ Assisted in data acquisition and detector configuration, enabling low-q speckle capture from particles ≤ 30 nm and recording promising SAXS and single-particle diffraction patterns.
- ♦ Collaborated with a multi-institutional team to coordinate beamline operations and document experimental results.

Data Analyst, MaRTy Fieldwork and Urban Data Collection, ASU

May 2024 - July 2024

- ♦ Directed urban fieldwork and collected MaRTy data for precise shade analysis, demonstrating dedication to thorough research.
- ♦ Applied advanced stats to validate MaRTy temperature data, confirming shade's urban heat mitigation.
- ♦ Transformed MaRTy data into actionable insights, informing Shade Lab-led urban planning for sustainable heat resilience

PROJECTS

Reborn - Scientific Visualization Framework

June 2025 - Present

- ♦ Enhanced an open-source Python scientific computing framework for processing, visualizing, and analyzing X-ray diffraction detector data from X-ray Free Electron Laser (XFEL) experiments used in structural biology and ultrafast imaging research.
- ♦ Extended the Qt-based PADView interface by improving detector ring annotation, labeling, and interactive visualizations.
- ♦ Diagnosed and resolved cross-environment compatibility, dependency, and rendering issues involving PyQt/PySide scientific software stacks in Linux development environments.

ReddWall - Reddit Moderation Tool

May 2026 - June 2026

- ♦ Built a Reddit app to help moderators instantly lock and remove comment threads during raids, brigading, and spam bursts.
- ♦ Enabled bulk moderation across entire comment trees with keyword filtering to target only harmful or off-topic content.
- ♦ Added a preview mode and audit log so moderators can review impact before acting and track all moderation history.

Microjet Image Analysis for XFEL Sample Delivery

February 2026 - April 2026

- ♦ Developed Python tools to analyze XFEL imaging data for liquid microjet characterization in sample delivery experiments.
- ♦ Implemented image processing and cross-correlation algorithms to compute jet velocity, diameter, breakup behavior, and angle.
- ♦ Automated preprocessing, metadata extraction, and visualization workflows for nozzle benchmarking and experimental analysis.

Web-Based Q&A Platform – Full-Stack Java Application

Jan 2025 - May 2025

- ♦ Designed and implemented a multi-role Q&A system with CRUD operations and input validation using a Finite State Machine.
- ♦ Built features like private messaging, reviewer approval, and trust-based review filtering.
- ♦ Applied UML design, Javadoc documentation, and JUnit testing to ensure system reliability.

Graph Routing & Pathfinding Engine

May 2025 - June 2025

- ♦ Developed a graph-based routing engine to compute shortest paths between connected nodes using pathfinding algorithms.
- ♦ Implemented Dijkstra's algorithm with adjacency lists and min-heap optimization for scalable shortest path computation.
- ♦ Built a command-line interface for loading graph datasets, executing routing operations, and reconstructing traversal paths.

PUBLICATIONS

Ansari, A. et al. (incl. Divyanshu Tandon) "Synchronized droplet nozzle for in-vacuum X-ray scattering experiments," Lab on a Chip, Advance Article, 2025, DOI: [10.1039/D5LC00063G](https://doi.org/10.1039/D5LC00063G).